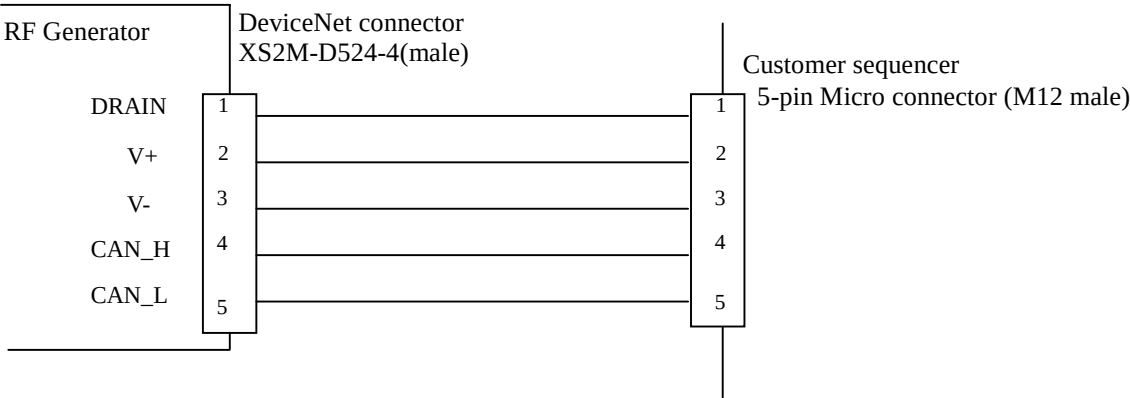


DeviceNet interface of RF Generator and Matchbox

Table 1. DeviceNet Assignment

Pin No	Signal	Signal type
1	DRAIN	Shield
2	V+	DeviceNet supply
3	V-	DeviceNet supply
4	CAN_H	Transmission & Reception data
5	CAN_L	Transmission & Reception data



Device Profile

(1) General

Conforms to DeviceNet Specification	Volume 1 CIP, Release 3.2 Volume 3 DeviceNet, Release 1.4
Vendor Name	KYOSAN ELECTRIC MFG.
Device Profile	Generic

(2) Physical Conformance Data

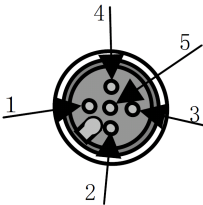
Network Power Consumption	DC24V MAX: 55mA
Connector Style	Sealed-Micro
Isolated Physical Layer	Yes
MAC ID Setting	Software Selectable
Communication Rate Setting	Auto Detection
Communication Rates Supported	125Kbit/s, 250Kbit/s, 500Kbit/s

(3) Communication data

Pre-defined Master / Slave Connection Set Group Only 2 Server	
Dynamic Connection (UCMM)	No
Explicit Message Fragmentation	Yes

(4) Connector and signal

PinNo	Name
1	DRAIN
2	V+
3	V-
4	CAN_H
5	CAN_L



(5) I/O Polling Message: MASTER -> RF

Offset	7 (MSB)	6	5	4	3	2	1	0 (LSB)
0	Forward Power Out (Low byte)							
1	Forward Power Out (High byte)							
2	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)
3	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)
4	(Reserved)	(Reserved)	(Reserved)	(Reserved)	Remote out	RST	(Reserved)	On
5	Soft Start Rise Time (Low byte)							
6	Soft Start Rise Time(High byte)							
7	Soft Start Fall Time(Low byte)							
8	Soft Start Fall Time(High byte)							
9	Level Arc Detection(Low byte)							
10	Level Arc Detection(High byte)							
11	Arcing Count(Low byte)							
12	Arcing Count(High byte)							
13	Restart Time(Low byte)							
14	Restart Time(High byte)							
15	Recover Soft Start							
16	Recognize Count(Low byte)							
17	Recognize Count(High byte)							
18	Signal Mask Time							
19	Signal Mask Time(Delay)							
20	Signal Output Time(Delay)							

Forward Power Out	Power Output setting , Unit:1(W)	
Soft Start Rise Time	Ramp up time from zero to rated output power. Rise time: 0.0-6000.0(s), Unit:0.1(s)	
Soft Start Fall Time	Ramp down time from rated output power to zero. Fall time: 0.0-6000.0(s),Unit:0.1(s)	
Level Arc Detection	: 0.00-10.00 or limited figure by internal setting (V), Unit:0.01(V)	
Arcing Count	: 0-9999(count), Unit:1(count)	
Restart Time	: 1-5000(ms), Unit:1(ms)	
Recover Soft Start	: 0.0-25.5(s), Unit:0.1(s)	
Recognize Count	: 0-500(count), Unit:1(count)	
Signal Mask Time	: 0.0-25.0(s), Unit:0.1(s)	
Signal Mask Time (Delay)	: 0.0-5.0(s), Unit:0.1(s)	
Signal Output Time (Delay)	: 0.0-5.0(s), Unit:0.1(s)	
Offset-bit	Name	State
4-0	On	Power ON 0 = Power Output OFF / 1 = Power Output ON
4-2	RST	Alarm RESET 0 = Reset OFF / 1 = Reset ON
4-3	Remote out	When RF generator is changed into remote (=DeviceNet) mode, this bit is changed into 1 from 0. Note: In the case of the following conditions, this bit is ignored. Power output is on During user menu operation Something alarm(s) occur(s)

I/O Response Message: RF -> MASTER

Offset	7 (MSB)	6	5	4	3	2	1	0 (LSB)
0	Forward Power Out (Low byte)							
1	Forward Power Out (High byte)							
2	Forward Power In (Low byte)							
3	Forward Power In (High byte)							
4	Reflected Power In (Low byte)							
5	Reflected Power In (High byte)							
6	Soft Start Rise Time (Low byte)							
7	Soft Start Rise Time(High byte)							
8	Soft Start Fall Time(Low byte)							
9	Soft Start Fall Time(High byte)							
10	Level Arc Detection(Low byte)							
11	Level Arc Detection(High byte)							
12	Arcing Count(Low byte)							
13	Arcing Count(High byte)							
14	Restart Time(Low byte)							
15	Restart Time(High byte)							
16	Recover Soft Start							
17	Recognize Count(Low byte)							
18	Recognize Count(High byte)							
19	Signal Mask Time							
20	Signal Mask Time(Delay)							
21	Signal Output Time(Delay)							
22	Arcing	DC Shut Down	Abort	AC Sag	Remote In	Failure	RF Off	RF On
23	(Reserved)	Interlock	Slave Not Ready	FWD Limit	OV Limit	OC Limit	Vpp Limit	REF Limit
24	(Reserved)	(Reserved)	DC failure	Amp imbalance	Cover open	Blown fuse	Tempera- ture rise	Fan failure
25	OV limit protection	OC limit protection	Vpp limit protection	REF limit protection	DC shut down	DC source fail (OV)	FWD limit protection	Arcing count
26	Control abort (H)	Control abort (L)	(Reserved)	(Reserved)	(Reserved)	(Reserved)	Over Vpp protection	Over REF protection
27	(Reserved)	(Reserved)	Interlock (RF-ON)	(Reserved)	Water shortage	S-cont failure	Chopper failure	MB Fail
28	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)	Water leakage	(Reserved)	(Reserved)

Forward Power Out	Report Output Power setting, Unit:1(W)
Forward Power In	Forward Power monitor, Unit: 1(W)
Reflected Power In	Reflected Power monitor, Unit: 1(W)
Soft Start Rise Time	Report, Unit:0.1(s)
Soft Start Fall Time	Report, Unit:0.1(s)
Level Arc Detection	Report, Unit:0.01(V)
Arcing Count	Report, Unit:1(count)
Restart Time	Report, Unit:1(ms)
Recover Soft Start	Report, Unit:0.1(s)
Recognize Count	Report, Unit:1(count)
Signal Mask Time	Report, Unit:0.1(s)
Signal Mask Time (Delay)	Report, Unit:0.1(s)
Signal Output Time (Delay)	Report, Unit:0.1(s)

Offset-bit	Name	State
22-0	RF On	When RF is ON, “1” is set for this bit.
22-1	RF Off	When RF is OFF, “1” is set for this bit. Note: When coincidence of RF ON flag and RF OFF flag continues, the communication error is occurring.
22-2	Failure	When RF is failure, “1” is set for this bit. Please get attribute no. 162,163 or 164 for detail.
22-3	Remote In	When RF is DeviceNet mode, “1” is set for this bit.
22-4	AC Sag	When AC input power failure “1” is set for this bit.
22-5	Abort	When control impossible caused by output abnormal “1” is set for this bit.
22-6	DC Shut Down	When DC unit failure or low or excess input voltage, “1” is set for this bit.
22-7	Arcing	When arcing occurs, “1” is set for this bit.
23-0	REF Limit	When RF is droop control by reflected power limit, “1” is set for this bit.
23-1	Vpp Limit	When RF is droop control by Vpp limit, “1” is set for this bit.
23-2	OC Limit	When RF is droop control by output current limit, “1” is set for this bit.
23-3	OV Limit	When RF is droop control by output voltage limit, “1” is set for this bit.
23-4	FWD Limit	When RF is droop control by forward power limit, “1” is set for this bit.
23-5	Slave Not Ready	When RF slave is not ready, “1” is set for this bit.
23-6	Interlock	When interlock contact is opened, “1” is set for this bit.
24-0	Fan failure	When the cooling fan failed, “1” is set for this bit.
24-1	Temperature rise (over heat)	When the temperature in RF generator is too high, “1” is set for this bit.
24-2	Blown fuse	When the fuse is blown, “1” is set for this bit.
24-3	Cover open	When the cover opened, “1” is set for this bit.
24-4	Amp imbalance	When the RF amps are imbalance, “1” is set for this bit.
24-5	DC failure	When the DC section failed or instantaneous failure,”1” is set for this bit.
25-0	Arcing count	When arcing occurs more often than setting,”1” is set for this bit.
25-1, 25-4, 25-5, 25-6, 25-7	Limit protection	When overload (FWD, REF, Vpp, OC, and OV.) continues longer than set time,”1” is set for this bit.
25-2	DC source fail (OV)	When DC unit failed, ”1” is set for this bit.
25-3	DC shut down	When repeated more than a specified number of the DC over-current, ”1” is set for this bit.
26-0	Over REF protection	When Reflected power excess continues longer than setting, ”1” is set for this bit.
26-1	Over Vpp protection	VPP over voltage continues longer than setting, ”1” is set for this bit.
26-6, 26-7	Control abort (H) / (L)	When the RF control impossible caused by output abnormal,”1” is set for this bit.
27-0	MB fail	When something alarm(s) on the matcher occur(s),”1” is set for this bit.
27-1	Chopper failure	When chopper unit failed,”1” is set for this bit.
27-2	S-cont failure	When Power supply of the middle-stage amplifier failed,”1” is set for this bit.

27-3	Water shortage	When low cooling water flow (*.* liters or below), "1" is set for this bit.
27-5	Interlock (RF-ON)	When interlock contact is opened due to an external factor, "1" is set for this bit.
28-2	Water leakage	When water leakage was detected, "1" is set for this bit.

(5) I/O Polling Message: MASTER -> Matchbox

Offset	7 (MSB)	6	5	4	3	2	1	0 (LSB)
0	Match Manual Out (USINT)							
1	Tune Manual Out (USINT)							
2	(Reserved)	(Reserved)	(Reserved)	(Reserved)	Remote Out	RST	Move	Auto / Manual

Match / Tune Manual Out	Motor position setting with manual 0 to 99, Unit: 1(%) note: The value of 100 or more is disregarded
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Offset-bit	Name	State
2-0	Auto / Manual	0=Auto matching Control / 1=Manual operation
2-1	Move	Manual operation 0=DISABLE / 1= enable
2-2	RST	Alarm Reset 0=Reset OFF / 1=Reset ON
2-3	Remote Out	When Matchbox is changed into remote (=DeviceNet) mode, this bit is changed into 1 from 0. Note: In the case of the following conditions, this bit is ignored. While in user menu operation Something alarm(s) occur(s)

Combination of flags

Move	Auto / Manual	operation
(no care)	0: Auto	Auto matching Control
0: disable	1: Manual	All motor stopping
1: enable	1: Manual	Motors move to settings (Manual out)

(6) I/O Response Message: Matchbox-> MASTER

Offset	7 (MSB)	6	5	4	3	2	1	0 (LSB)
0	Match Position In (USINT)							
1	Tune Position In (USINT)							
2	VDC In (Low byte)							
3	VDC In (High byte)							
4	Vpp In (Low byte)							
5	Vpp In (High byte)							
6	Water flow In (Low byte)							
7	Water flow In (High byte)							
8	Match CW	Match CCW	Tune CW	Tune CCW	Remote In	Failure	RF ON	Auto / Manual
9	Match CW Limit	Match CCW Limit	Tune CW Limit	Tune CCW Limit	(Reserved)	(Reserved)	(Reserved)	(Reserved)
10	(Reserved)	(Reserved)	(Reserved)	(Reserved)	(Reserved)	Recall timeout	Hardware Limit	(Reserved)
11	(Reserved)	(Reserved)	Water leakage 2	Water leakage 1	Cover open	(Reserved)	Water shortage	Fan failure

Match Position In	Match Position Monitor 0 to 99 , Unit: (%)
Tune Position In	Tune Position Monitor 0 to 99 , Unit: (%)
VDC In	VDC Monitor , Unit:1 (V) (-4000 to 4000 = 0xF060 to 0x0FA0)
Vpp In	Vpp Monitor , Unit:1 (V)
Water flow In	Water flow Monitor , Unit:0.01 (Liter/minute)

Offset-bit	Name	State
8-0	Auto / Manual	0=Auto matching control 1=Manual operation
8-1	RF ON	0=RF OFF 1=RF ON
8-2	Failure	0=no failure detected 1= failure detected. Please get attribute no. 165 for detail.
8-3	Remote In	0=Local 1=Remote (DeviceNet mode)
8-4, 8-5, 8-6, 8-7	Match CW, Match CCW, Tune CW, Tune CCW	0=stop 1= It moves in the direction.
9-4, 9-5, 9-6, 9-7	Match CW Limit, Match CCW Limit, Tune CW Limit, Tune CCW Limit	0= Normal 1= At the limit position
10-1	Hardware limit	When mechanical limit was detected, "1" is set for this bit.
10-2	Recall timeout	When some motors did not reach the position in time, "1" is set for this bit.
11-0	Fan failure	When the cooling fan failed, "1" is set for this bit.
11-1	Water shortage	When low cooling water flow (*.* litters or below), "1" is set for this bit.
11-3	Cover open	When the cover opened, "1" is set for this bit.
11-4	Water leakage 1	When water leakage (1) was detected, "1" is set for this bit.
11-5	Water leakage 2	When water leakage (2) was detected, "1" is set for this bit. (option)

Request Matchbox to Manual Operation

